**GOOGLE DATA ANALYSIS CERTIFICATE**

**CASE NUMBER 2**

· **How can a wellness Technology company play it smart?**

## 1. Ask Phase

· What topic are you exploring?

Wellness and health data from a database.

· What is the problem you are trying to solve?

Trends and behaviour in wellness data that use health smart devices and then with these insights make recommendations about how the bellabeat smart devices should perform and how they could ameliorate.

· How could these trends help influence Bellabeat marketing strategy?

These trends will be a useful guide to be assertive as to how users utilize the company's healthcare products in order to be more precise, user-friendly, more empathy towards customers and to satisfy in a better manner the client's product needs.

· How can your insights drive business decisions?

We can be more accurate and focus on women's desires and needs, projecting an image of a company that really cares and supports women, using health data to leverage our continued expansion market making profitable decisions that result in a win-win market strategy.

· What metrics will you use to measure your data to achieve your objective? Who are the stakeholders?

It depends on the data available, but a first guess would be health topics women are mostly centered or worried about.

Apps functionalities that are the most commonly checked by women (number of clicks and taps could be a metric).

Any surveys if available about how well the smart devices performed.

Visible and strong trends that reveal how women use their smart devices to consider applying this knowledge.

· Who is your audience for this analysis and how does this affect your analysis process and presentation?

The audience is the executive team. They will mostly care about results, trends, and the story data. Maybe they will not expect comprehensive and detailed story data, but they will check if there is consistency in the story and processing of data in my report, along with a structured logical thinking to come up with reliable conclusions and recommendations.

· How will this data help your stakeholders make decisions?

This data will convince the stakeholders to make directions about the product functionalities and performance to gain better insights in how the current company's products have to change to attract more customers and strengthen the current client's loyalty to the brand.

- Key tasks

Identify the business task

**Business Task: Deliver high-level recommendations about womens smart device data choices and trends in order to inform about Bellabeat marketing strategy.**

Key stakeholders:

- Executive team

**After reviewing data**: There is no description for the activities that each participant did throughout the month, only a scale used to classify intensity, and these classifications to discriminate categories are not explained in the archives. Also, there is not enough description and records of body mass, height, kg, and other basic measurements to consider when detailing fitness profiles, not even the gender can be determined. Nevertheless, at least there is a daily total count of lost calories and data about manually logged in sessions. This will reduce the scope of the analysis to only discovering general trends of activity for participants. Needless to say, specific profiles will not be ascertained.

For instance, there is a lack of data for weight loss, BMI, Fat, and Weight, there are only 67 records, and 59 for weight and BMI that belong to two people, out of the 33 that make up the total sample. So there is not enough data for all the population.

## 2. Prepare phase

- Data location

Data located with a CC0 code (Public domain, data openness) within a database titled : Fitbit Fitness Tracker Data on the Kaggle webpage through user Mobius.

Datasets generated by respondents to a distributed survey via Amazon Mechanical Turk between 03-12-2016 and 05-12-2016 dates. Thirty eligible users consented to the submission of personal Tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring.

- Data organization

The data is structured and organized by columns and rows. All formats of the archives are in csv allowing them to be exported in Excel. Nevertheless, there are certain datasets that are very difficult to handle in Excel, because of numerous rows that exceed Excel's capacity to perform well. There are certain measures for minutes that come in a wide and long range, making it more easier to handle in Excel, but still it is better to opt for SQL or R to transform data for this big data.

- Issues with credibility and bias.

Of course there are issues with bias and credibility. The data does not tell if the participants are women or men. Then, the metrics to categorize women and give recommendations for women will be greatly biased. The data quality to make specific profiles is not far away from being random. This is the element that mostly affects credibility in this study.

- Using ROCCC (Reliable, O) method:

  Is data reliable?

  After checking data formats and sources it can be considered reliable but incomplete.

- Is data original?

  No, it is data from an Open Source, from a second-party data, it comes originally from Fitabest. Second-party data can be more reliable than third-party data, but still, after a close examination of data, it is evident the lack of data to even outline specific profiles for each participant.

- Is data Comprehensive?

  Not all data is comprehensive, the profiles of the different Id's are not complete, consequently determining sexes and weight loss data will be impossible to achieve based on missing data and how weight loss cannot be traced or determined thoroughly with just one month of data.  Only the loss of calories can be determined.

  Nonetheless, the data for minutes, hours and days for intensity, steps and calories is complete and would provide great insights about their trends.

In addition, there is only data about a month for all participants. The variation of a year seasonally will not give annual trends, then this limitation of data will not provide a comprehensive understanding of fitness trends of people.

- Is data Current?

  It can be considered current for our analysis. The impact of a time gap from 3 to 4 years is not relevant, therefore we have a good source of information based on current data.

- Is data Cited?

  It is from a reliable organization that has many apps and gadgets to measure physical activity via an Amazon survey Mechanical Turk. We can rely on this data to perform analysis, based on the source of data. Also, it comes from an open source and is open for debate. Many comments have been posted in Kaggle to evaluate its source, totality and validity. It also has a score of 10 out of 10 on the kaggle website.

- Addressing licensing, privacy, security, and accessibility

  As it is an open data source, its accessibility is easy and it has no restrictions of privacy. Only accessibility and privacy are factors to be considered while calculating and obtaining results to avoid data alteration through data transformation.

- How did you verify the data's integrity?

  Sample bias would be important if only a single record of activity per day, per minute or per second, for the month the data was produced. But, this is not the case, daily activities almost amount to one thousand. Seconds and hours have more than one million records. Using ROCC to rate the quality of data, it is considered a good source of information.

- Are there any problems with the data?

  The datasets metadata described 30 participants, but counting the total identification numbers these added up to 33. Maybe there were some devices that malfunctioned and were replaced for the same participant, or the metadata was erroneous.

## 3. Process phase

- Tools used

  Excel for all daily activities and to modify some data from the hourly datasets. R is chosen based on its comprehensive and high quality detailed visualizations and also because it is perfect to use on big data for hourly data.

- Data integrity

  Affirmative, by using ROCCC factors to ensure data integrity and credibility.

- Steps taken to ensure that data is clean

  Checking the format for all values to make sense. Searching to find the fitabase pdf guide which opportunely gives a brief description of the categories and metric used by each device. Although, the formula for classification for intensities and distances is not provided.

  Also, when summarizing, sorting and filtering data, findings about duplicates and values that make no sense were considered outliers and were deleted.

- How can you verify that your data is clean and ready to analyze?

  Apart from the lack of data to constitute profiles for each participant, during the cleaning process, the totals and categories for classification for many metrics were not equal. That means, that the parts did not amount to the total when these were classified into diverse categories.

  Tracker distance is not equivalent for all cases to the total distance. Only calculations were performed with the total distance.

  On the other hand, the total of minutes throughout the day is 1440. In spite of this widely known conversion number, the total number of minutes for all categories is not consistent for all 940 records. There are values that are less or even have more seconds per day, that means, that this data had to be considered isolated and incomparable, the total from its categories (Highly active, Fairly active, Lightly active and sedentary from the total of minutes). Unfortunately the categories did not add up to the right value.

  Also, sleep day minutes analysis revealed only data for 24 Id numbers and total number of participants is 33 for the entire month of this study, that would indicate that the device only recorded data from 24 participants. It is impossible that the sleep time for 9 participants was 0, implying they did not sleep at all. Sleep analysis was set aside as inconsistent data and was not taken into account when calculating the total number of minutes per day. This last decision was fully supported when adding the sleep minutes with the extra categories of intensity; there were some records that were higher than the 1440 minutes per day.

  Also, illogical records were spotted. For instance, there are some values of total distance verification that had negative values higher than 100 meters.These data needed to be deleted as these are outliers, another example of inconsistent data is all data for which the total number of steps was zero for a given day,  but had under the sedentary minutes category a value less than 1440.

In conclusion, data is inaccurate, data did not amount to right totals, so a comprehensive analysis can be done individually, but comparison of categories with the whole, that is total daily activities with total sum of daily classifications of data will not be precise.

- Documentation of cleaning process to review and share results

  Daily activity and hourly activity is recorded in Google Sheets by the historical record functionality and R Console and history options provide a thorough sequence of all changes for data transformations. In next sections, code will be provided about the summarizing and

## 4. Analyze and Share Phase

People don't have conscious habits about these devices, only 32 records have manually logged total distance out of 940 records of daily activities.

As only the total count of records for sleep minutes for all ids were 412, but the total records for track distance, and total distance added up to 912 each, it is highly improbable that in average half the population only slept more than each two days. So all minutes of sleep were disregarded when doing analysis for total minutes.

When the cleaning process was achieved satisfactorily, the results of classified intensity, total number of steps, total distance and total numbers of minutes for the month per day od the week for all participants are given below:

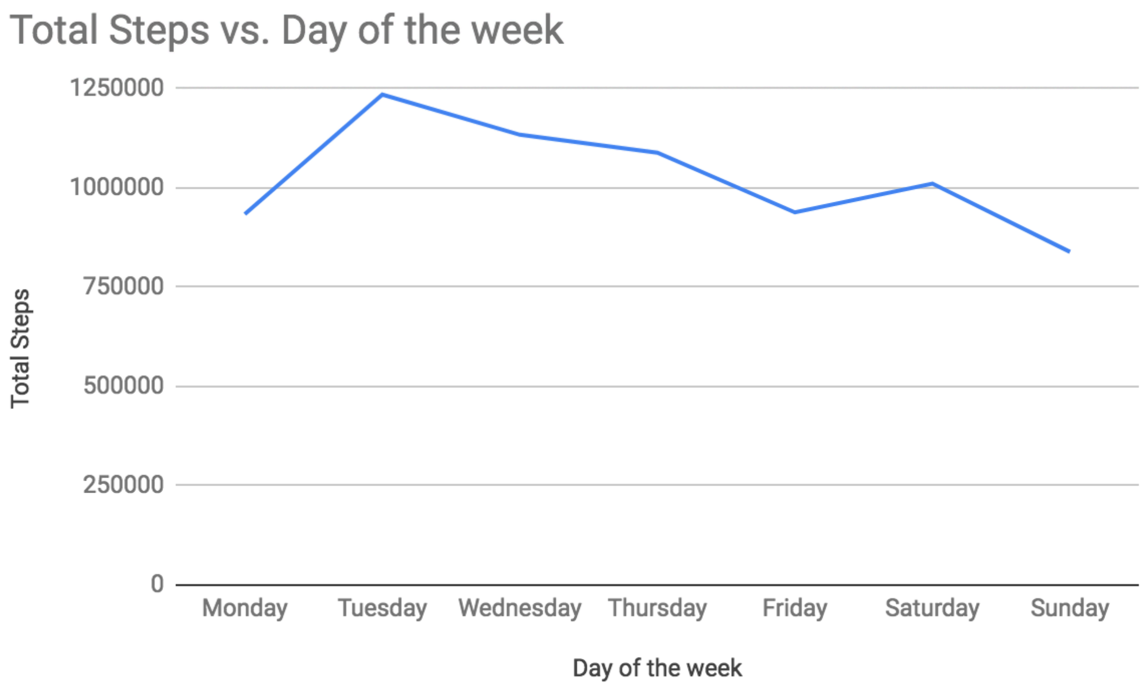

*Figure 1. Total steps vs. Day of the week*

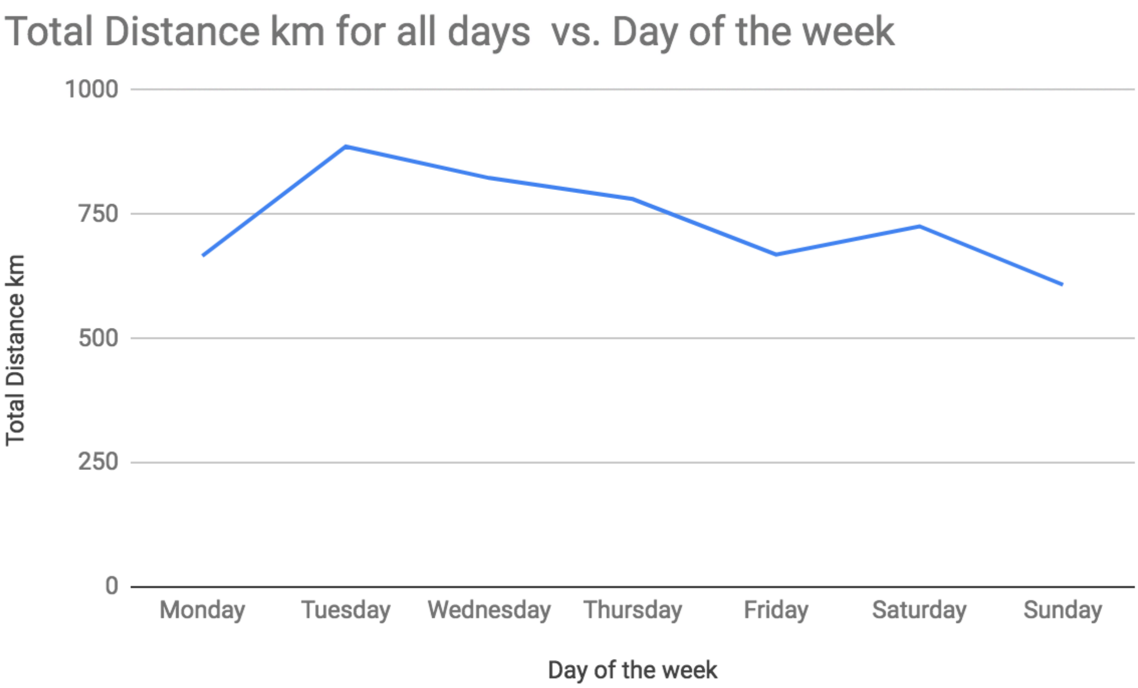


*Figure 2. Total distance vs Day of the week*

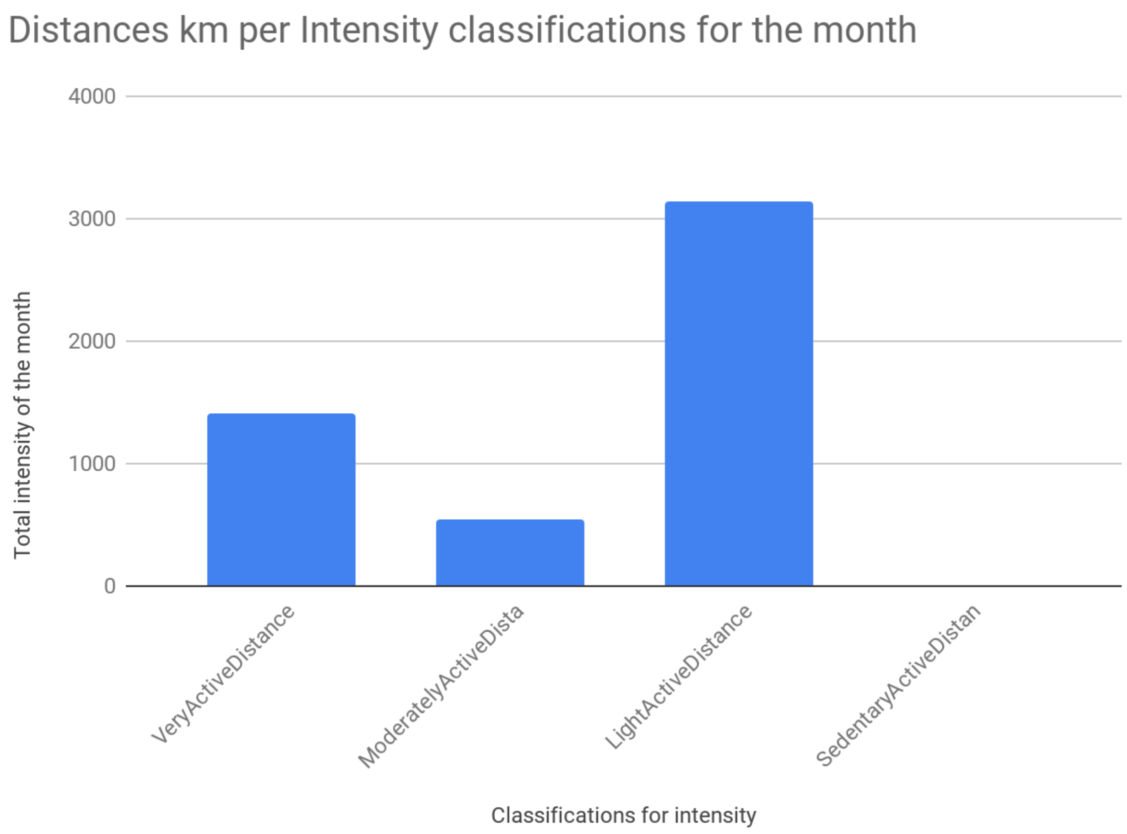


*Figure 3. Distances per intensity for the month.*

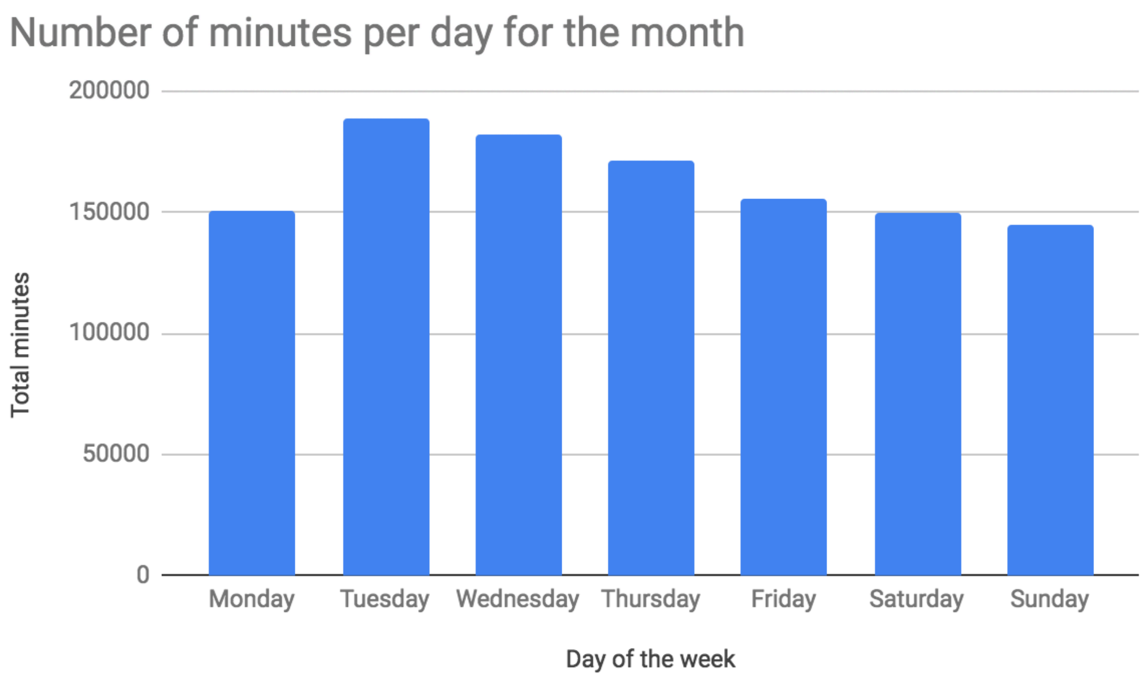


*Figure 4. Number of minutes per day for the month.*

Based on the four graphics given above, people were more active on Tuesdays, which is not what is first thought out as a common pattern for daily exercise. It was expected that days with more intensity, number of steps and calories were on Saturdays and Sundays. Perhaps if people were more involved in the dataset these trends will eventually change.

Also, when calculating percentages of distance traveled by intensity, it is clear that sedentary distance was minimal, while light intensity has the most percentage for all participants, making it a total over 60% of the distance traveled.

Given the fact that total calories per day lost was given for each participant, we wanted to see variations of loss of calories per day according to different categories of minutely intensity activity. These graphics are presented below:

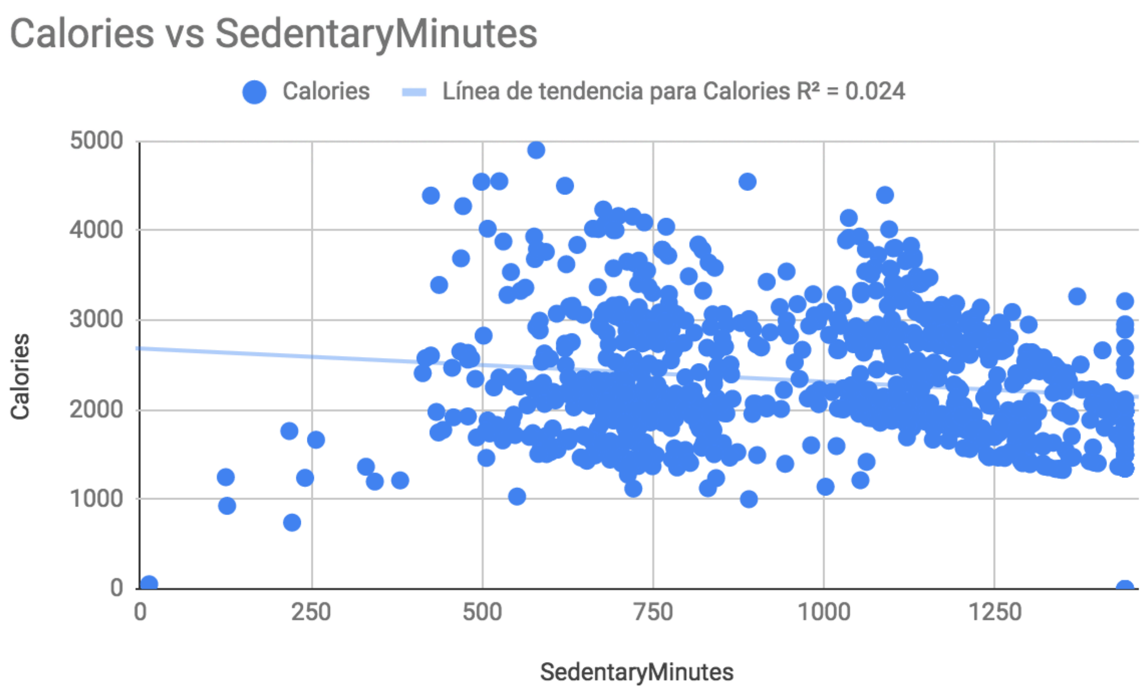


*Figure 5. Calories vs. Sedentary minutes*

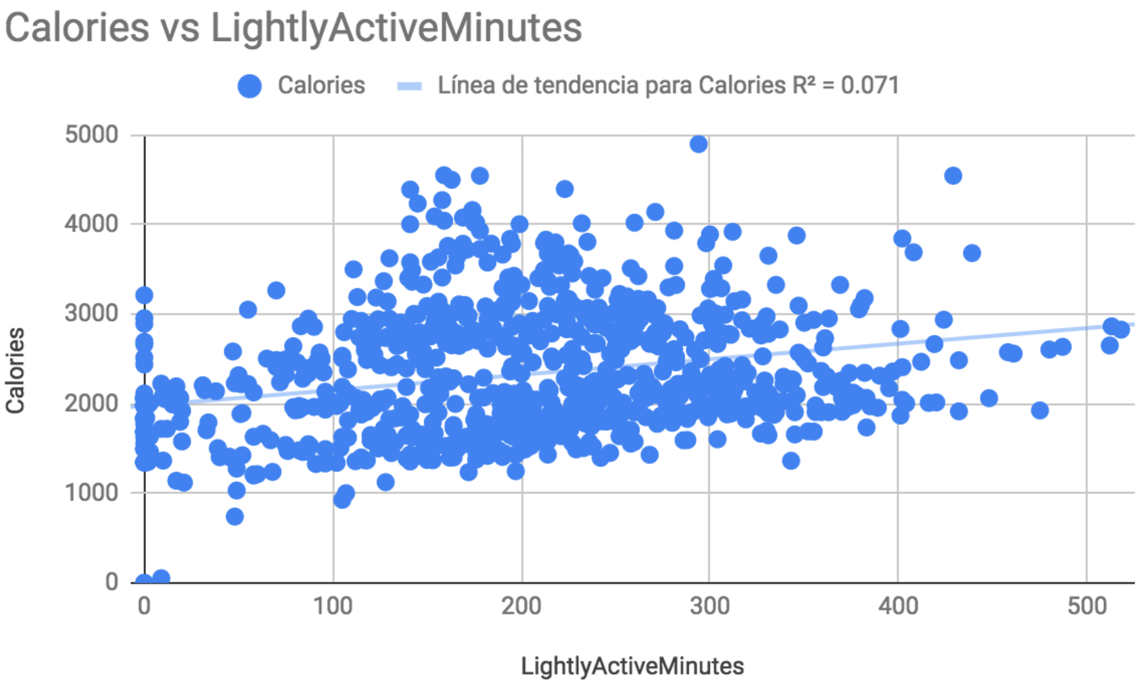


*Figure 6. Calories vs Lightly active minutes*

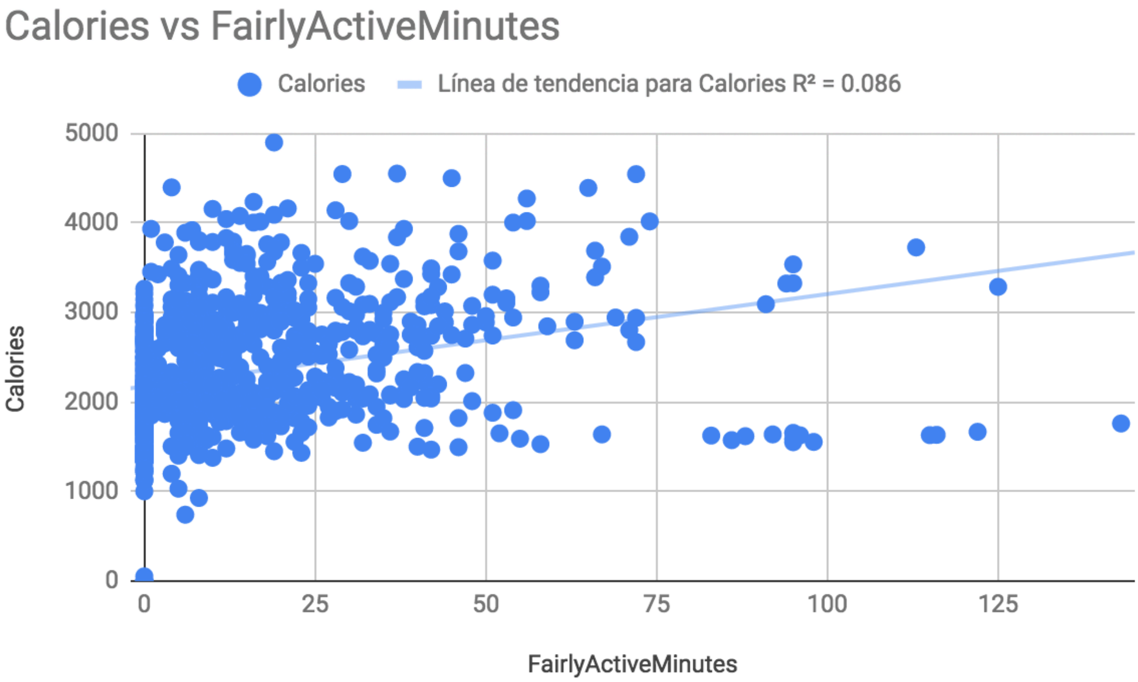


*Figure 7. Calories vs Fairly active minutes*

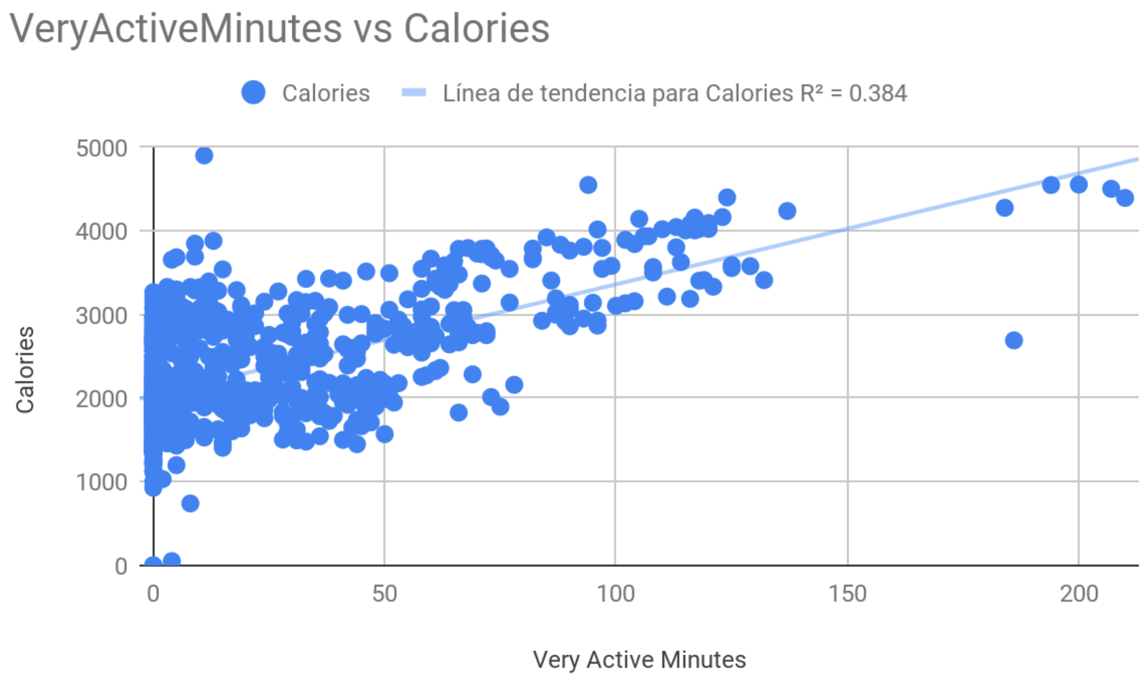


*Figure 8. Very active minutes vs calories*

It can be clearly seen that sedentary minutes have an inverse linear correlation with calories lost. On the contrary, the very active minutes category correlates the most linearly with calories lost, followed by fairly active minutes. Alternatively, an additional graph of total active minutes excluding sedentary minutes was performed:

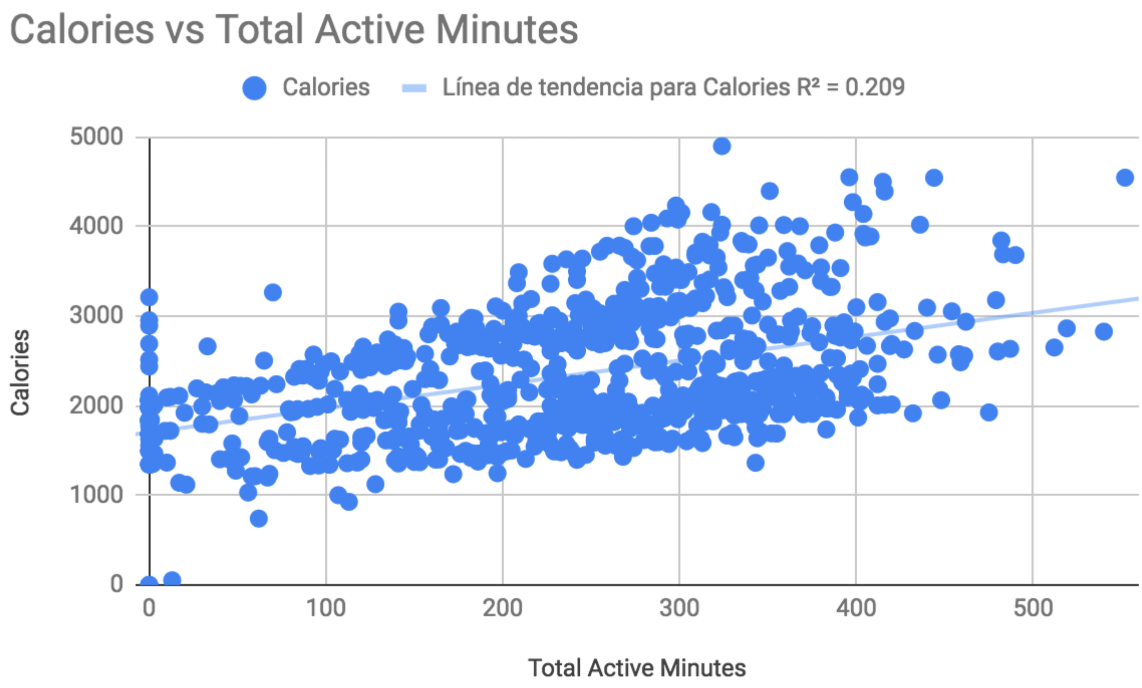


*Figure 9. Calories vs Total Active Minutes categories*

Also, Viewing figure 9 it can be that the straight line meets with the x-axis at approximately 1700 calories, where a total of zero minutes of active intensity is recorded. So this should be considered an average of calories loss by absolutely doing nothing. Weight loss then happens when calories lost per day is over this number.

Besides, comparing the linear coefficient correlation (R) results for the total of active minutes with only the highly active minutes category, this last has a major coefficient, so it can be concluded that the category that leads to a greater loss of calories is for those activities with the greatest intensity.

We also wanted to assess the number of total minutes under the category of sedentary activity versus those among lightly and very active minutes categories for all participants throughout the month:

| Category | Total | Percentage |
|---|---|---|
| **Very Active Minutes** | 19895 | 1.74% |
| **Fairly Active Minutes** | 12751 | 1.11% |
| **Lightly ActiveMinutes** | 181238 | 15.85% |
| **Sedentary Minutes** | 929900 | 81.30% |

*Table 1. Minutes classification and percentages*

It can be seen that sedentary minutes make up the most of the daily total minutes, excluding sleeping minutes, although sedentary activities make a very slight distance according to figure 3.

## REMAINING DATA:

### MET AND CALORIES, INTENSITY AND STEPS FOR MINUTES

Given the fact that daily and hourly data was consistent in the results, it was not necessary to perform calculations minutely.

If we should give indications for people on what they should do and when, then the minute time frame is not possible. In addition, the aggregation of minutely data will result in hourly data, data that was already cleaned and transformed to achieve results.

### HEARTBEAT EACH 5 SECONDS:

The case should be focused not in finding that the more MET, and the greater the heartbeat the more calories are lost. This previous statement is obvious and has been widely studied. It is not a result that leads to action, instead, variation about most active hours was perceived as a great feature to be studied.

In addition, for heartbeat data to be satisfyingly studied, more information should be provided, including diet, and key basic metrics like: list of activities performed daily, related heart diseases, and diseases and conditions that might affect the heart beating ratio.

### HOURLY DATA

Hourly data was cleaned and tested to get results. This data was treated in Excel as well as in R. Hourly data was considered valuable to establish which hours throughout the day were the most active in order to see visible patterns.

The coding to summarize and provide graphics for hourly sets in R is provided below:

```
> install.packages("tidyverse")
>library(tidyr)
>library(dplyr)
>HC <- read.csv("Hourly Calories.csv")
>View(HC)
>HC1 <- HC %>%
group_by(Hour) %>%
summarize (sumcal = sum(Calories))
>View(HC1)
>HC3=read.csv("Hourly Calories 24 hr format - HC.csv")
>HC3$datetime <- as.POSIXct(paste0("2021-11-21 ", HC3$Hour), tz = "GMT")
>ggplot(HC3, aes(x = datetime, y = sumcal)) +geom_line(color = "dark blue")+scale_x_datetime(date_labels = "%H:%M", date_breaks = "1 hour")+theme(axis.text.x = element_text (angle=45))+xlab("Hour")+ylab("Total Calories")+ggtitle("Total Calories Throughout the day")+geom_point(color="Red")
>HI<- read.csv("hourlyIntensities_merged.csv")
>View(HI)
>HI1 <- HI %>%
>group_by(Hourly.Time) %>%
summarize (SumTotalIntensity = sum(TotalIntensity), SumAverageintensity = sum(AverageIntensity))
>View(HI1)
>x = read.csv("Hourly_Intensity_Cleaned - Hourly Intensity Cleaned.csv") >
>x$datetime <- as.POSIXct(paste0("2016-09-23", x$Hourly.Time), tz = "GMT")
> ggplot(x, aes(x = datetime, y = SumAverageintensity)) +geom_line(color = "dark blue")+scale_x_datetime(date_labels = "%H:%M", date_breaks = "1 hour")+theme(axis.text.x = element_text (angle=45))+xlab("Hour")+ylab("Average Intensity")+ggtitle("Average Intensity Throughout the day")+geom_point(color="Red")
> ggplot(x, aes(x = datetime, y = SumTotalIntensity)) +geom_line(color = "dark blue")+scale_x_datetime(date_labels = "%H:%M", date_breaks = "1 hour")+theme(axis.text.x = element_text (angle=45))+xlab("Hour")+ylab("Total Intensity")+ggtitle("Total Intensity Throughout the day")+geom_point(color="Red")
```

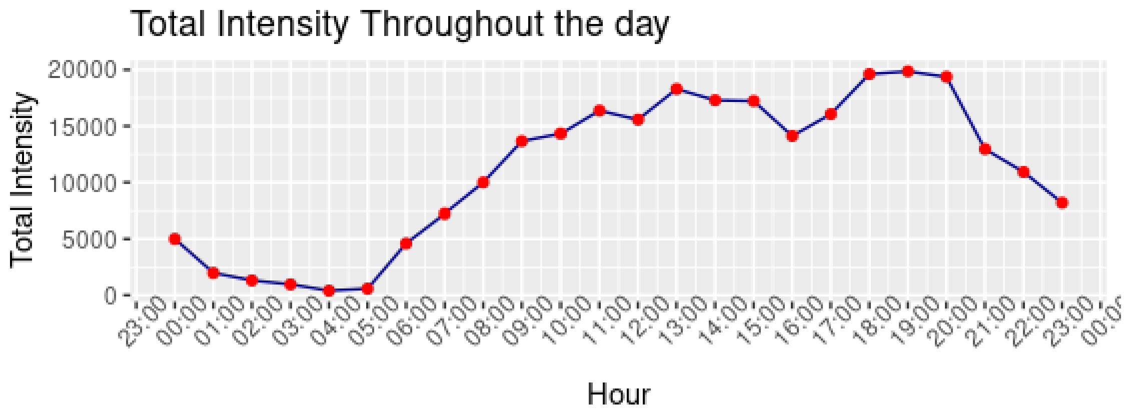


*Figure 10. Total Intensity throughout all days*

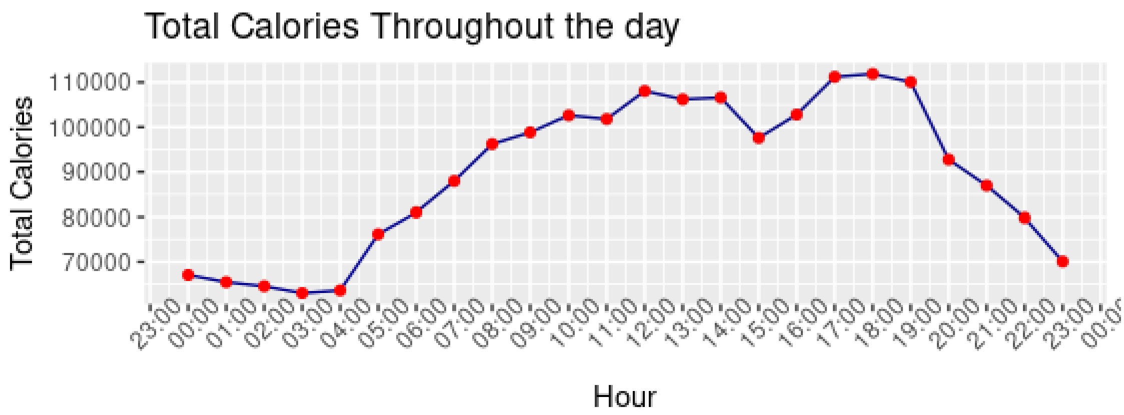


*Figure 11. Total Calories Throughout all days*

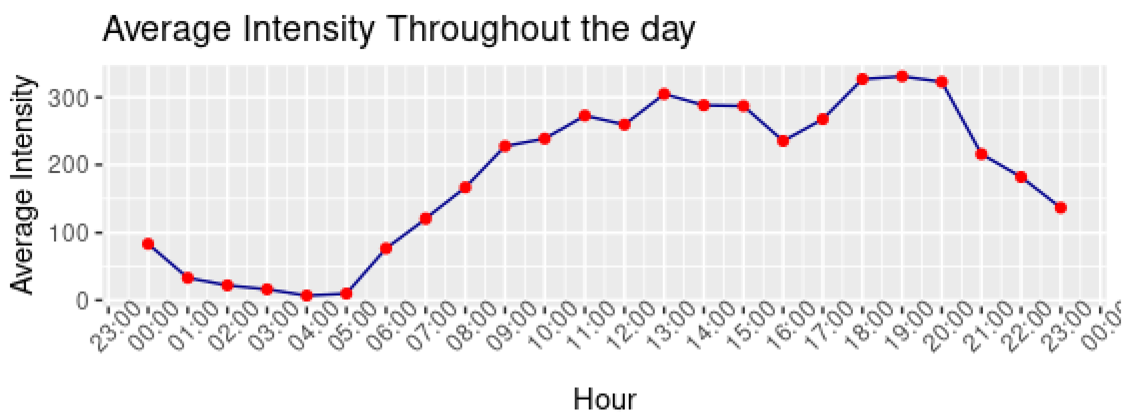


*Figure 12. Figure 11. Average Intensities Throughout all days*

When it comes to hourly trends, all measurements of average intensity, total intensity and total calories are similar throughout time, all graphs look alike. Based on the Plots and calculations performed per hour for all days and participants, the hours that calories are most lost are very noticeable.

The trend is logical and can be described with most intensity hours happening at daytime and at night drastically lowering, with the greatest loss of calories in the range of 06:00 P.M and 08:00 P.M. There's also a peak between 01:00 p.m and 03:00 P.M

If we want to send notifications and reminders to people in order to encourage them to stay healthy, the best hours should be between 09:00 a.m and 08:00 p.m, specifically in the hours with the most activity, that is at 11:00 a.m, the peak from 01:00 pm and 3:00 p.m and last but not least, the peak between 06:00 p.m and 08:00 p.m

**Indications about data :**

- More studies should be performed in order to gain insights about specific fitness behavior for women. To empower the female population we have to demonstrate our measurements and indications that ameliorate women's health shall be provided with enough data about their lifestyle.

- Data was not enough to carry out a comprehensive study of fitness people trends, the inconsistency, the lack of integrity for data, and limitations about data classification for people' profiles including gender, age, country, and weight affects the reliability and accuracy of the results.

- Data for just one month could be limited. As the data for countries was not procured and the timeframe was so short, it was impossible to determine the seasonal variations of exercise for these participants. Further studies should be conducted in order to determine variations and tendencies of physical activity throughout the year.

- It is convenient and reasonable to provide from this dataset only general metrics of data when solving the business task.

- Daily Sleep minutes were neglected because they are potentially misleading, the records per day are less than half for the total steps, total distance and calories categories per day. Additionally, sleep data for 9 Ids were missing. And the total number of minutes per day for some days were above 1440 when adding this category into the intensity groups per day.

**Results**:

- It is concerning that more than 80% of the total time in minutes for all participants were sedentary activities taking action.

- The records results are consistent (the more the distance and steps the greater the calories lost), and tendency for steps, minutes and total distance for every day of the week are strongly similar. Nevertheless, as previously stated one drawback from this set of data is the integrity of total records vs individual groups of data. Not all data matched the totals when reviewing data separated into categories.

- It is worth noting that the high intensity category linearly correlates more than the total number of minutes with high intensity, fair intensity and light intensity, joined together, this could only mean that this segment of intensity is the most convenient when the goal of exercising is to lose calories. The higher the intensity, the more calories are lost.

- Surprisingly, the overall total of minutes, distance and steps were the highest on Tuesdays, followed by Wednesdays, and then continuously decreased until Monday. These trend could change if there were enough data for more participants and for a longer period of time than just one month.

## 5. Act Phase

**What should be aimed at considering Marketing strategies and improvement of wellness devices?**

- Smart devices should measure activities independently for people. People expect for these devices to continuously record data without the customer's intervention -32 out of 940 records show manually logged in actions-. Even so, these devices should provide options of what should be tracked, adding customization and diverse options to customers. Based on this data set, we should focus on giving a message of automation but also companionship between customers and the diverse products Bellabeat manufactures.

- People are not very attentive to their fitness devices, thus, these devices should send notifications and reminders about how exercise is beneficial and leads to great health. Although the Bellabeat membership offers personalized guidance, the devices should promote good health with basic and general statistics and studies, focusing on why sedentary activity is harmful to welfare.

- Given the fact that people are very sedentary but lead very busy lives, the best choice is to promote messages telling them that <u>it is better to go hard even if it is for a short amount of time than to remain in the light and sedentary categories of intensity for long periods of time.</u> We should promote that effort and consistency are key when losing weight while showing the dismal results for sedentary people.

- Given the sedentary disposition for most participants, tracking devices should allow to have reminders and notifications for the hours of great activity, so as to encourage physical activity, and give them the boost to start doing more exercise. The idea is to deliver more promotion of physical activity with the aim to reduce sedentary and lightly intensity , switching to fairly and highly intensity categories so people will see an actual loss of weight, so as to produce efficient results, attracting and winning more customers in the long run

- It is greatly advisable to be more inclined about finding insights of people that really use the apps from the company, rather than data gathered in this database. There was no possible way to discriminate between male and female. Hence, indications about quantitative metrics should be treated carefully as one would expect some variation of data between male and female. Marketing strategies should be supported by exhaustive and accurate data for the female population.

***Thank you for viewing my first data case. Any feedback will be of great help.***